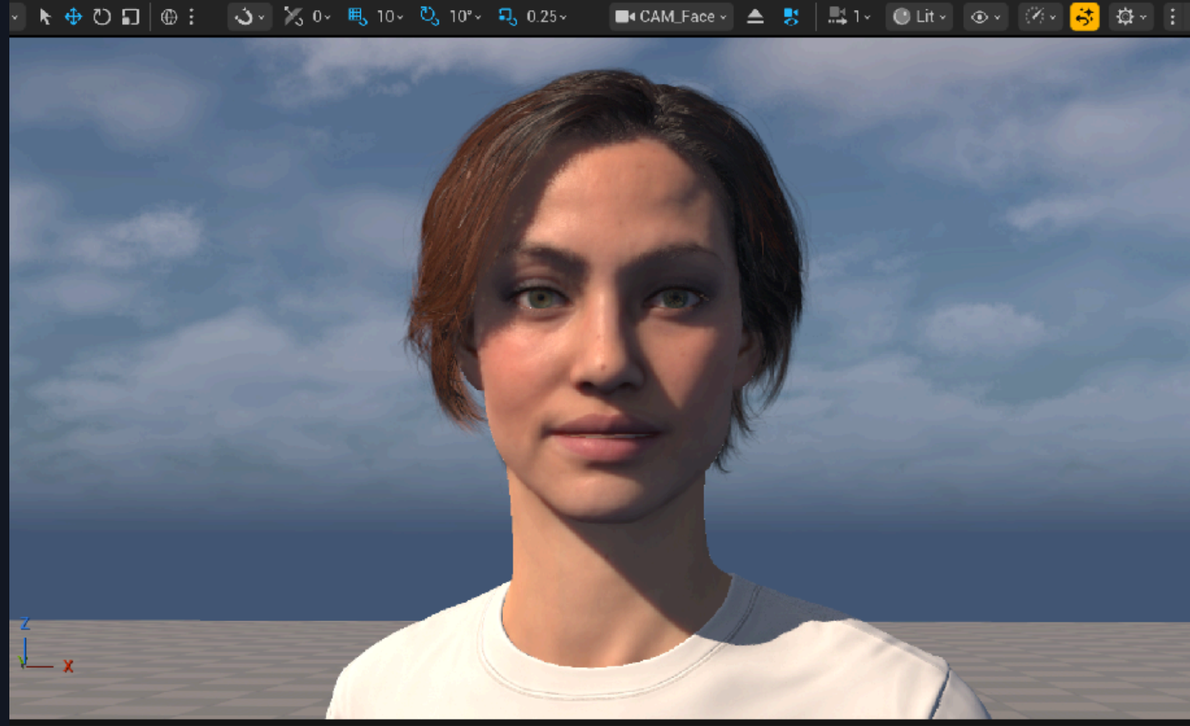
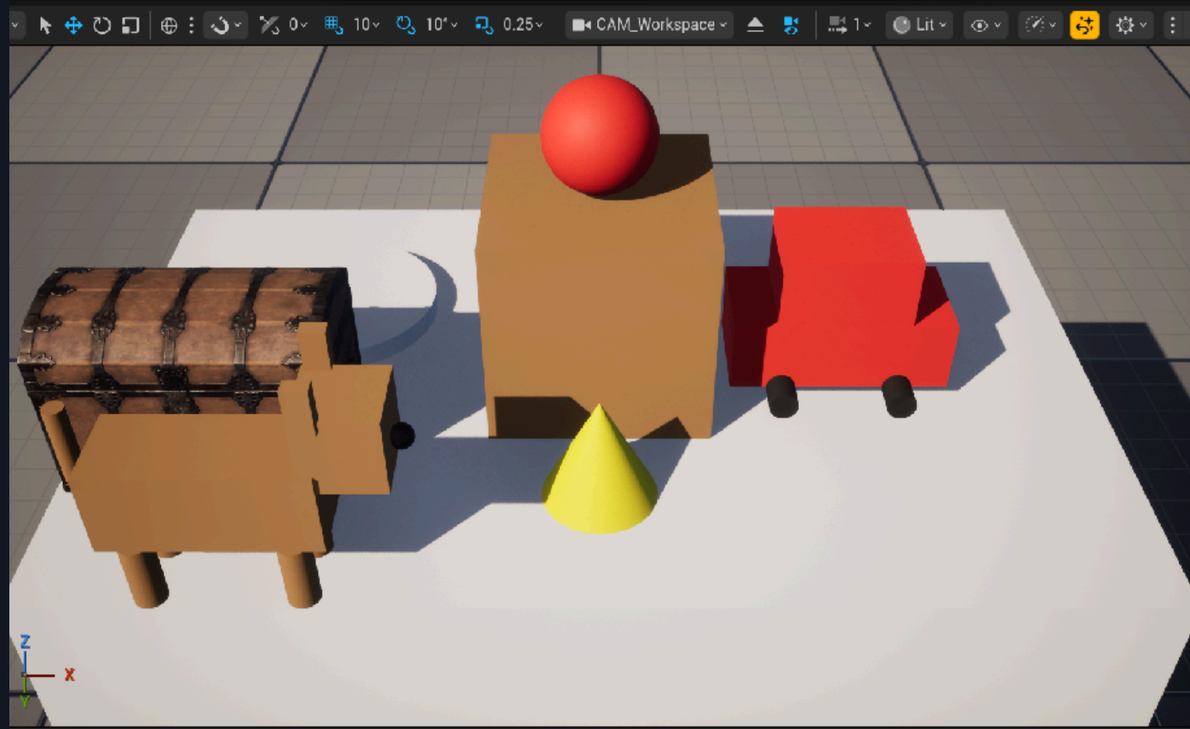
HACKATHON • 10 OCTOBER 2026

Meet Vivian

A talking game designer we built in one day

You speak. She answers with her voice and her face, and builds your ideas on a table.



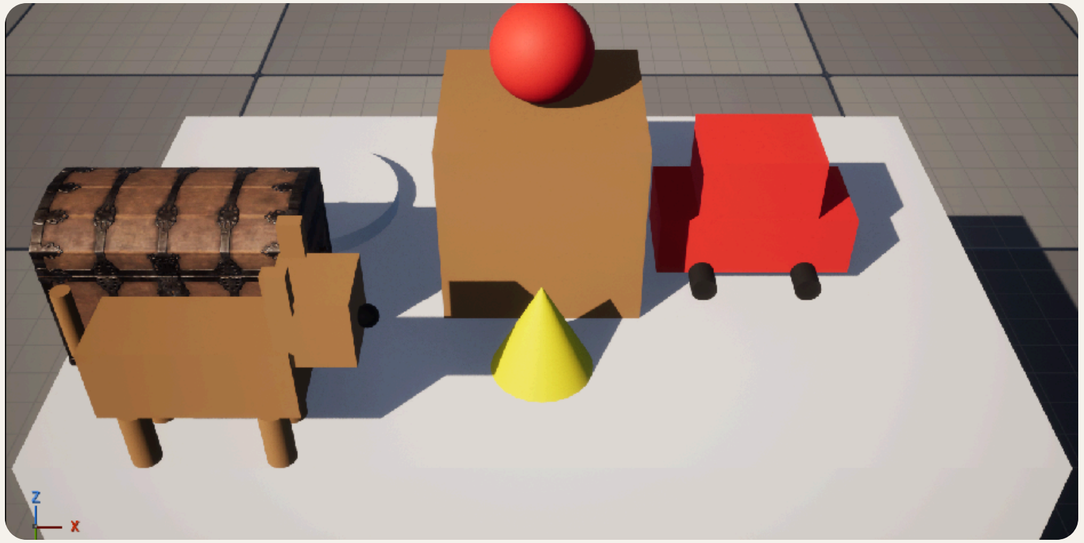


What Vivian can do



She talks and shows feelings

Speak into a mic and she answers out loud, quick and responsive. Her face shows 8 moods. She blinks, nods and looks you in the eye.



She builds what you ask for

On a second screen she places shapes, builds a dog or a car out of blocks, and brings in a real 3D treasure chest.

HOW IT WORKS

How one sentence travels



The Brain also writes hidden tags: [happy] sets her face, [[place red sphere on box]] runs Python inside Unreal and the table changes right away. You never hear the tags.

Our tech stack

Part	Tool	Its job, in simple words
3D character	Unreal Engine 5.8 + MetaHuman	Her realistic face, hair and body
Lip-sync	MetaHuman Audio Live Link	Turns her voice into mouth and face movement
Ears	Gradium speech-to-text	Writes down your words, knows when you stop
Brain	Google Gemini Flash-Lite	Writes her reply, her mood and table actions
Voice	Gradium text-to-speech	Reads the reply out loud in a natural voice
Audio pipe	VB-Audio Virtual Cable	A fake microphone that carries her voice to Unreal
Glue	Python bridge (asyncio, websockets)	Connects everything and streams audio live
Editor control	Unreal Python + Unreal MCP	Lets our code change the 3D scene and the editor
Body language	Unreal Animation Blueprints	Blinking, nodding, looking at the camera
Coding partner	Claude Code	Wrote, tested and debugged the code with us

The hard parts: feeling natural

Her voice came before her lips

Her face needs about half a second of sound before it can move her lips.

Fix: we measure that delay and hold back the voice you hear by the same time.

Answers felt slow

We waited for Gemini's whole reply before she started to speak.

Fix: streaming. She speaks each sentence as soon as it is written, so she feels responsive.

She cut people off

Faster "you are done talking" settings stopped long questions at a comma.

Fix: we kept the safer setting. Tip: say each line in one breath.

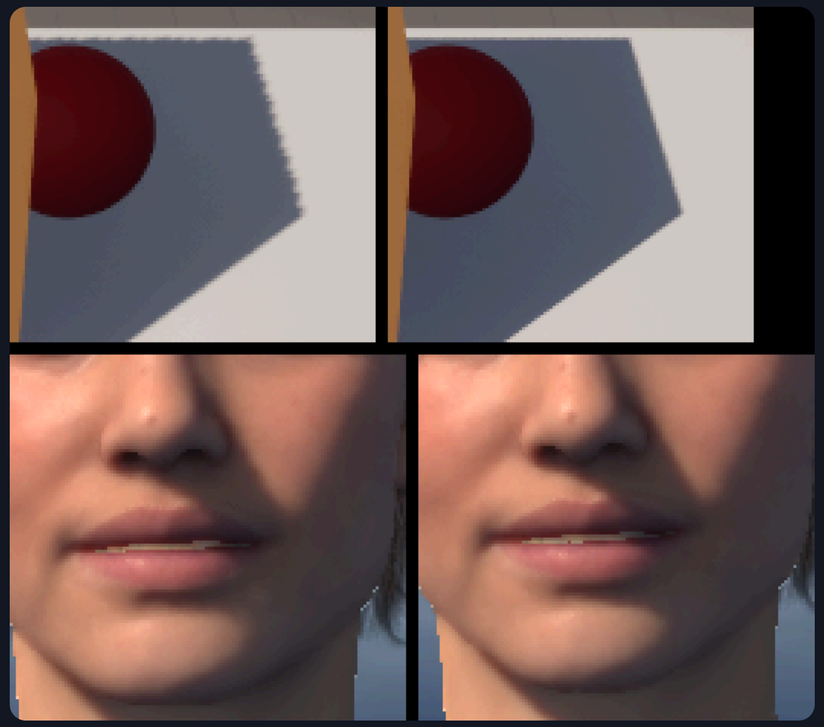
Her body jumped with every mood

The emotion animations started from a different pose, so she leaned in and swung back.

Fix: her body stays calm, and only her face shows the emotion.

Looking good on a split screen

BEFORE AFTER



Frozen picture

The new pane started with live updates switched off. Fix: turn Realtime on.

Flickering hair

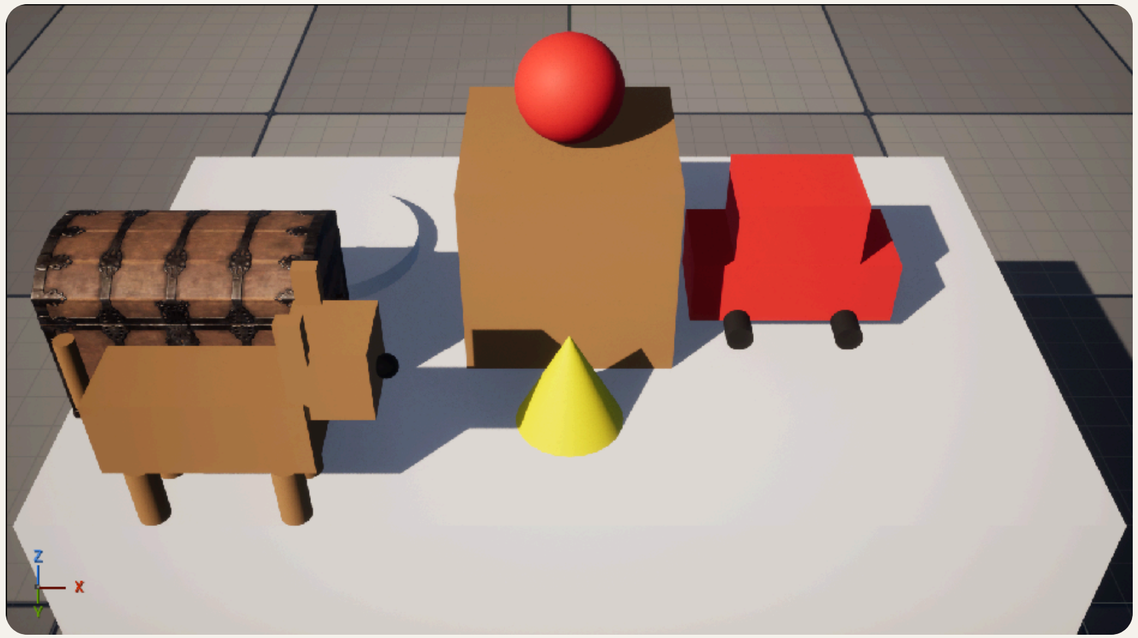
Each pane chose its own detail level, so her hair switched between two versions. Fix: always full detail.

Jagged shadows

Small panes made the shadows low-resolution, with saw-tooth edges. It was not Lumen. Fix: a sharper shadow setting.

BUILDING

She builds on the table



Gemini writes the tags, Python inside Unreal runs them right away, and you never hear them.

1. Simple shapes

`[[place red sphere on box]]`

Cubes, balls, cylinders and cones: on, left of, right of, in front of, behind.

2. Things made of blocks

`[[build small dog on table: …]]`

Up to 20 blocks, each with a size and a spot in cm. They move as one object.

3. A real game asset

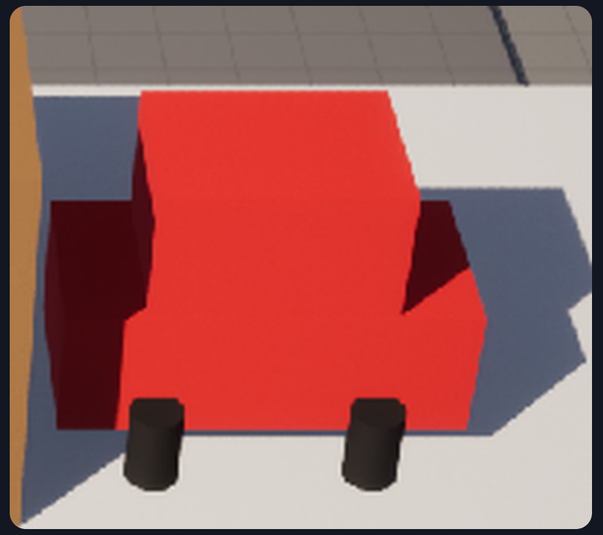
`[[place treasure chest …]]`

A free, photoscanned chest from Poly Haven (CC0), loaded as a glTF file.

BUILDING TOGETHER

Human in the loop

VIVIAN'S CAR



AFTER OUR FIX



The fix stays: the wheels are real parts attached to the car, so they move with it.

1. Vivian proposes

She designs a car out of blocks.

2. You spot a mistake

The wheels stand up like pegs.

3. You fix it by hand

Select the wheels in Unreal and set Rotation X to 90.

4. You keep going together

"Drive the car in front of the box": your fix moves with it.

What we learned, what's next

⚠️ Limits today

- You can't interrupt her yet.
- Lip-sync improved a lot, but we ran out of time to make it fully consistent.
- A pause of about a second ends your turn.
- Objects she designs herself vary in quality.
- She doesn't see the table; she only knows the list of objects.

💡 Next ideas

- Push-to-talk, and letting you interrupt her.
- More game-ready assets from Fab and Quixel.
- Let her see the table through a camera image.
- Turn table prototypes into playable game levels.

LIVE DEMO • PART 1 OF 3

Her feelings

#	You say	Her face
1	"Hi Vivian! Who are you, and what are you doing here today?"	happy
2	"Guess what, you were only created this morning. Today is your birthday!"	surprised
3	"I have some bad news. After the demo tonight, your server gets switched off."	sad
4	"Just kidding! We're keeping you. And the judges are watching you now."	excited
5	"Vivian, flurb the wobbly snorkel sideways, okay?"	confused
6	"Do you think you can design a better game than a human designer?"	confident
7	"Okay, tell me your best joke about game designers."	playful

Watch for: lips in sync with her voice, blinking, small nods, and her glances into the camera.

LIVE DEMO • PART 2 OF 3

Building on the table

#	You say	What happens
1	"Let's prototype a game level. What's on the table right now?"	She lists table, box, plate
2	"Put a red ball on top of the box."	A ball on the box
3	"Now add a blue cube to the left of the box and a yellow cone in front of it."	Two objects at once
4	"Move the red ball onto the plate."	The ball moves
5	"Our game needs a hero. Build a small dog in front of the plate."	A dog made of 10 blocks
6	"Put a green ball on the dog."	It rests on the dog's back
7	"Every hero needs a ride. Make a little red car to the right of the box."	A car, but its wheels stand upright

Each object appears right away, usually before she starts talking.

LIVE DEMO • PART 3 OF 3

Fixing it together, then finishing

#	You say	What happens
8	"Hmm, Vivian, the wheels are standing up like pegs. I'll fix them by hand."	She thanks you and waits
9	*Mute your mic. Select the 4 wheels, set Rotation X to 90, press Esc. Unmute.*	You fix her design by hand
10	"There, I turned the wheels the right way. What do you think?"	She is excited
11	"Now drive the car in front of the box."	The car moves with your fix
12	"Every game needs some loot. Put a treasure chest right of the box."	A real photoscanned chest
13	"Remove the blue cube."	The cube is gone
14	"So what game would you make with all this?"	She pitches a game idea
15	"Clear the table."	Back to table, box, plate
16	"Thank you Vivian. Say goodbye to the judges!"	A happy goodbye

Thank you!

Vivian was born today. She talks, feels and builds.

Ask her anything, or ask her to build your next game idea.



Q&A BACKUP • SPEED

How fast does she answer?

One real turn, timed from the moment you stop talking



Blue = waiting on purpose. Her face needs about half a second of sound to move her lips well, so we hold back the voice you hear by the same time.

Q&A BACKUP • SPEED

Speed in numbers

2.26 s

from the end of your sentence to hearing her (was 2.62 s before streaming)

1.3 s

to a written reply with Gemini Flash-Lite. The bigger Flash model took 3.3 s.

0.1 s

for her voice to start once the first sentence is written

0.2 s

to place or build an object on the table (0.16 to 0.36 s)

0.5–0.65 s

for her lips to follow the sound, so we delay the voice you hear to match

86–115 fps

smooth picture in the editor while she talks (RTX 3080 Ti)